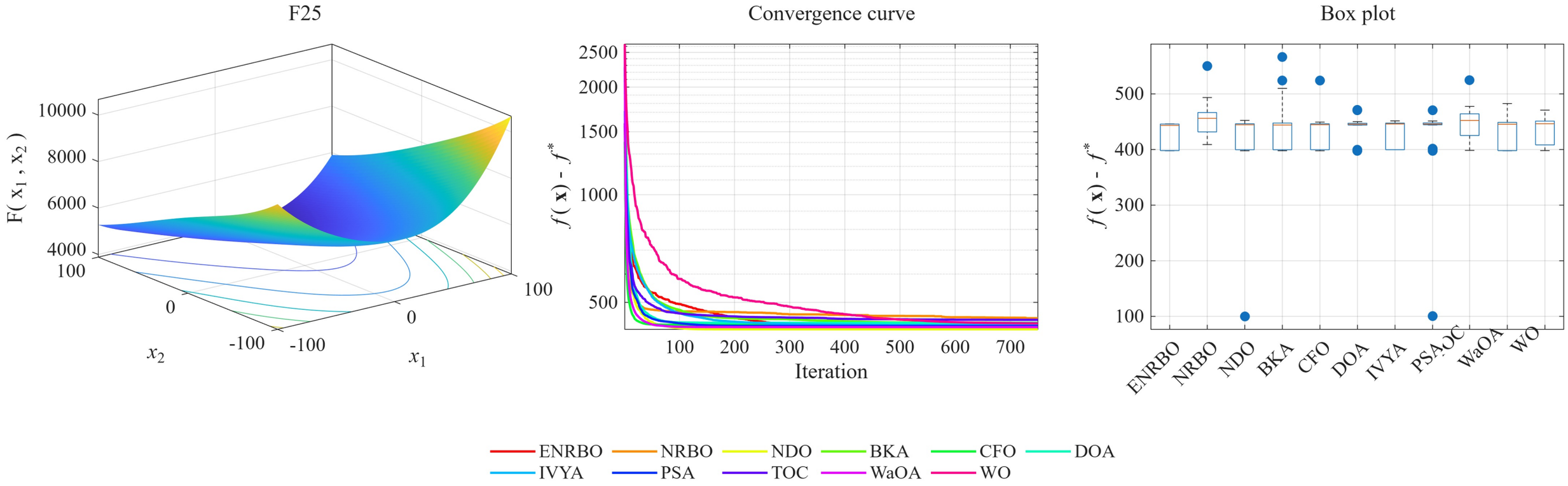

F25
F(x1 , x2)
10000
8000
6000
4000
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
2500
2000
1500
1000
500
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
500
400
300
200
100
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO